



# Introduction to Electromagnetic Spectrum (EMS)

112-BOLCEW02

# Concrete Experience

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- Who still uses their car radio?
- What setting do you use it on (FM or AM)?
- How do you switch between radio channels?
- What frequencies can you listen to?
- Why did we pick those frequencies?

# Terminal Learning Outcome



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The Student will define the characteristics of the Electromagnetic Spectrum (EMS) in a classroom environment, given a networked computer workstation, training aids, and appropriate references by demonstrating understanding of the EMS, frequency measurements, and spectrum division accurately to support the application of EW effects critical to the unit's mission, in accordance with EW Fundamentals Student Guide Vol. I, Navy's Electricity and Electronics Training Series (NEETS), and Training Circular (TC) Communications-Electronics Fundamentals 9-64, while achieving a minimum of 70% on all lesson Practical Exercises (PE) and end of module exam.

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# Learning Step Activities



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- Identify Wave Characteristics
- Identify basic concepts of the Electromagnetic Spectrum (EMS)
- Identify Radio Wave Frequency Bands within the RF Spectrum
- Identify the bands above Radio Waves and their typical uses

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# Waves Characteristics

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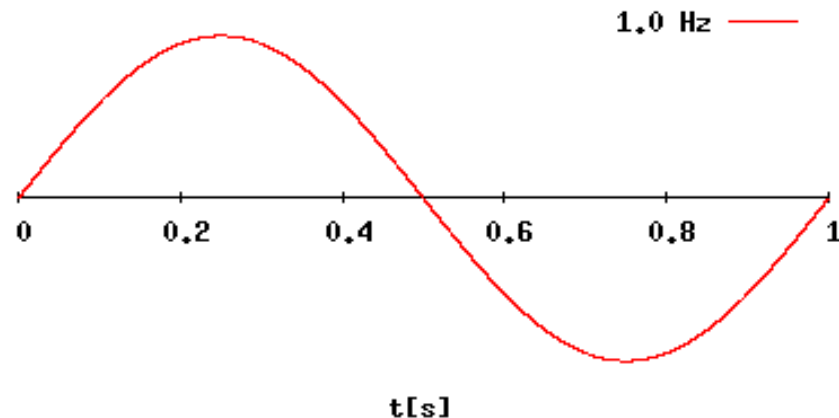
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# Waves Characteristics



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- **Radio signals are a form of Electromagnetic (EM) radiation.**
- **EM radiation propagates as a wave** and can be defined by standard wave characteristics.
- waves are typically generated by an alternating current.
- The standard unit to measure frequency is in Hertz (Hz).
- The frequency of a wave depends on the number of cycles completed in one second.



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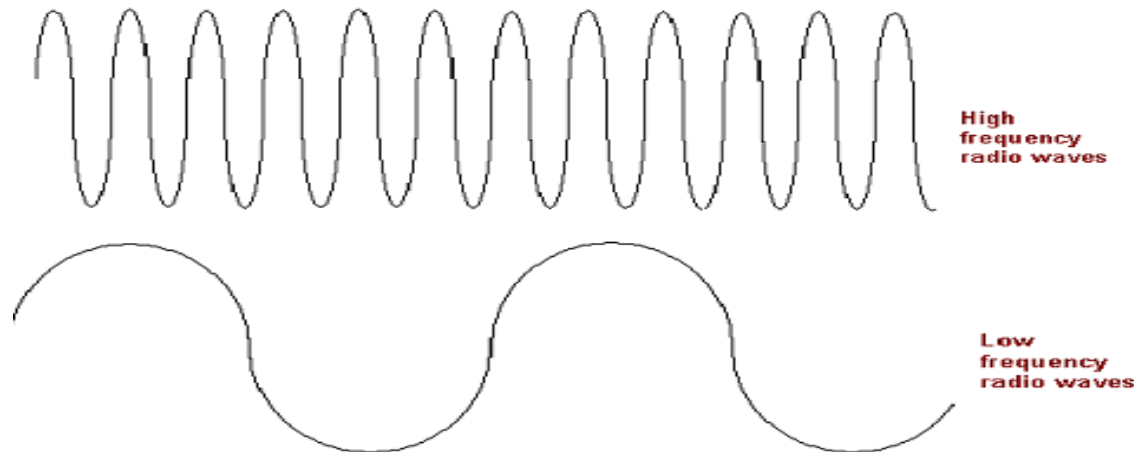
# Frequency



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Frequency is the number of waves or complete oscillations that pass a fixed place in a given amount of time.

In electronics, the time used to measure the frequency of a signal is 1 second.



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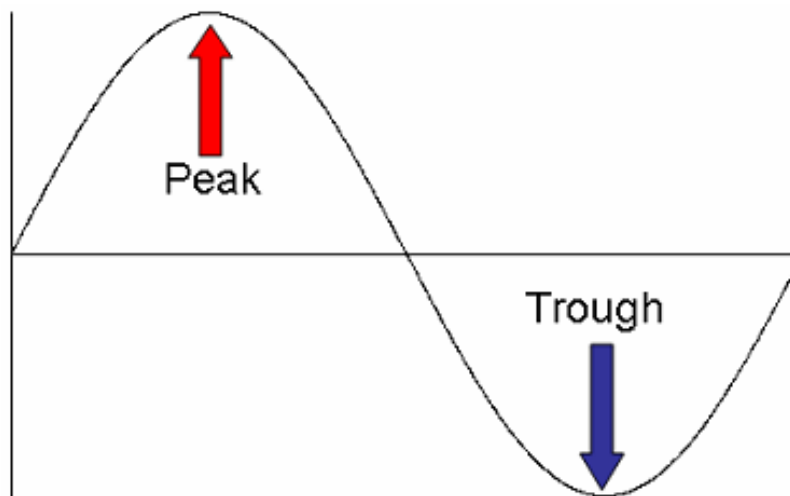
# Sine Wave



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The “peak” of a sine wave is the highest point on the wave (also known as the “crest”).

The trough of a sine wave is the lowest point of the wave.



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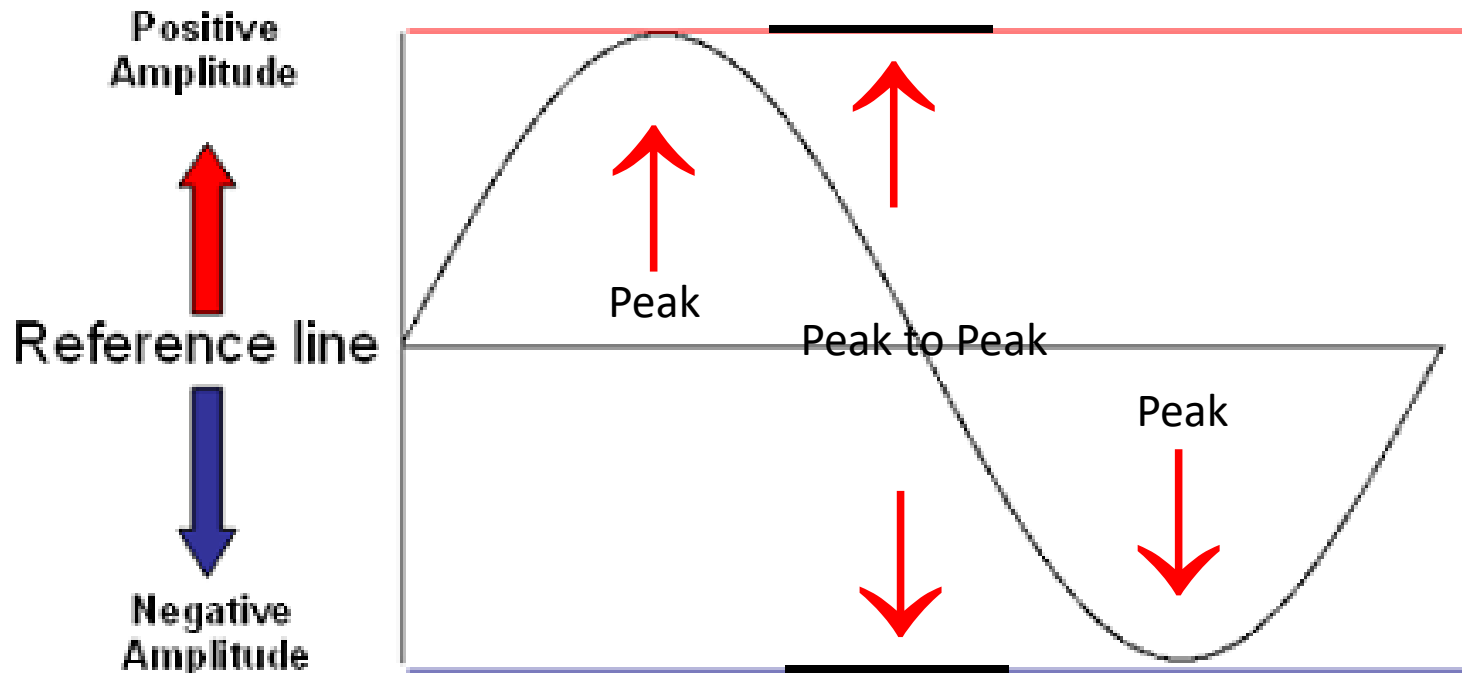
# Amplitude



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Amplitude is the height of a sine wave.

- Peak Amplitude - is measured from the reference line to the peak or the trough
- Peak-to-Peak Amplitude - is measured between the maximum positive to negative values



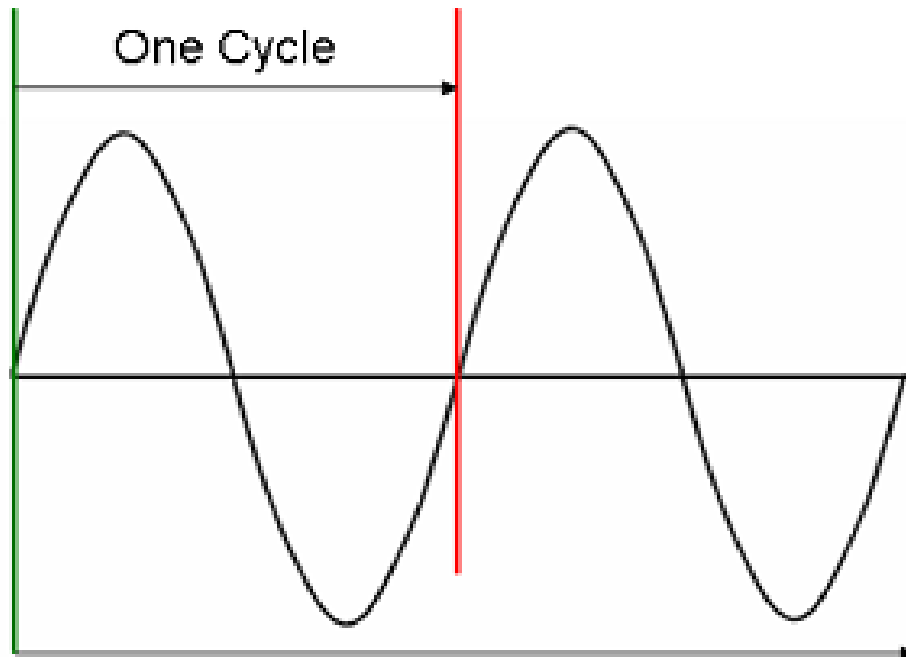
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# Cycle



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The cycle of a sine wave is the completion from any given point on a wave to the same point on the next wave.



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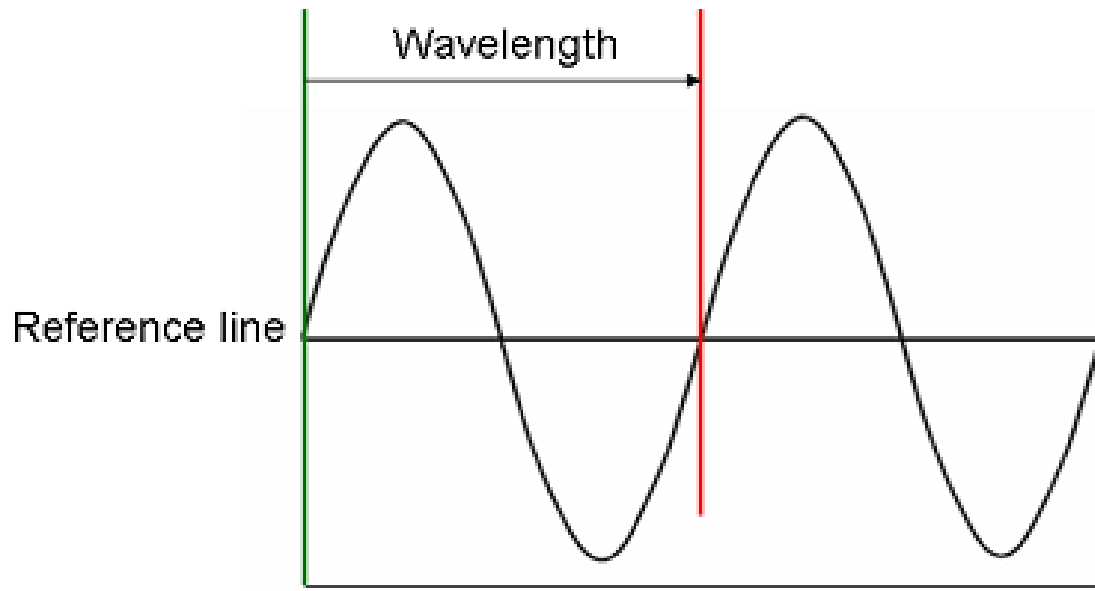
# Wavelength



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Wavelength is the distance between repeating units of a wave, usually measured in meters.

The lambda " $\lambda$ " symbol is used to represent wavelength.



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# Questions?

# Electromagnetic Spectrum (EMS) Concepts

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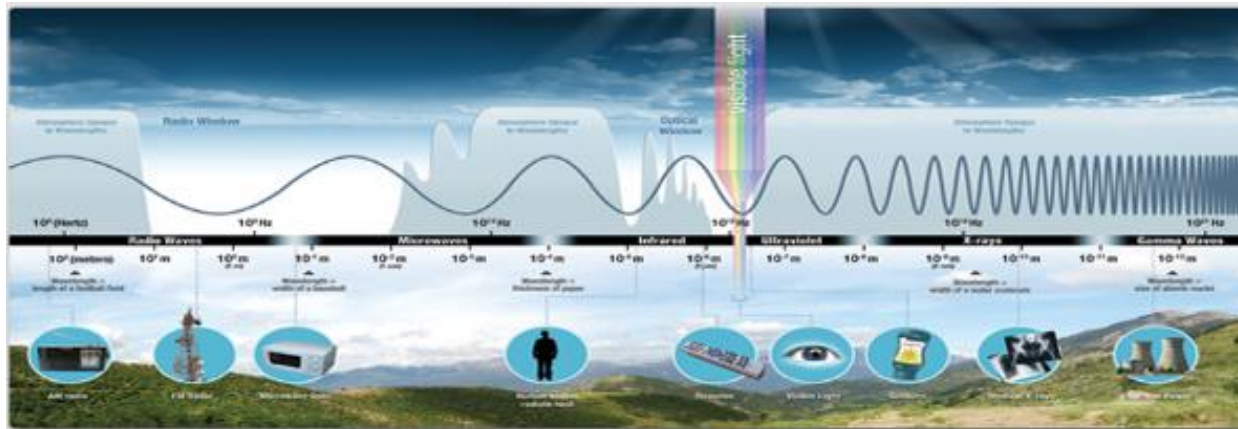
# Electromagnetic Spectrum Concepts



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The Electromagnetic Spectrum (EMS) is the range of all possible types of electromagnetic radiation (generally sorted by frequency).

Electromagnetic radiation travels in waves and spans a broad spectrum from very long radio waves with low frequencies to very short gamma rays with very high frequencies.



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# Radio Frequency



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Electromagnetic radiation is typically sorted and described by its frequency.

A Radio Frequency is :

- The number of oscillations of the perpendicular electric and magnetic fields per second is expressed in hertz
- measured in kilohertz, megahertz, or gigahertz; this is why radio dials are commonly labeled with kHz, MHz, and GHz

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# Radio Wavelength



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Some nongovernmental organizations categorizes radio waves by its wavelength.

A Radio wavelength is :

- The amount of distance covered by on complete cycle of the wave.
- measured in Meters (m), Centimeters (cm), or Millimeters (mm);

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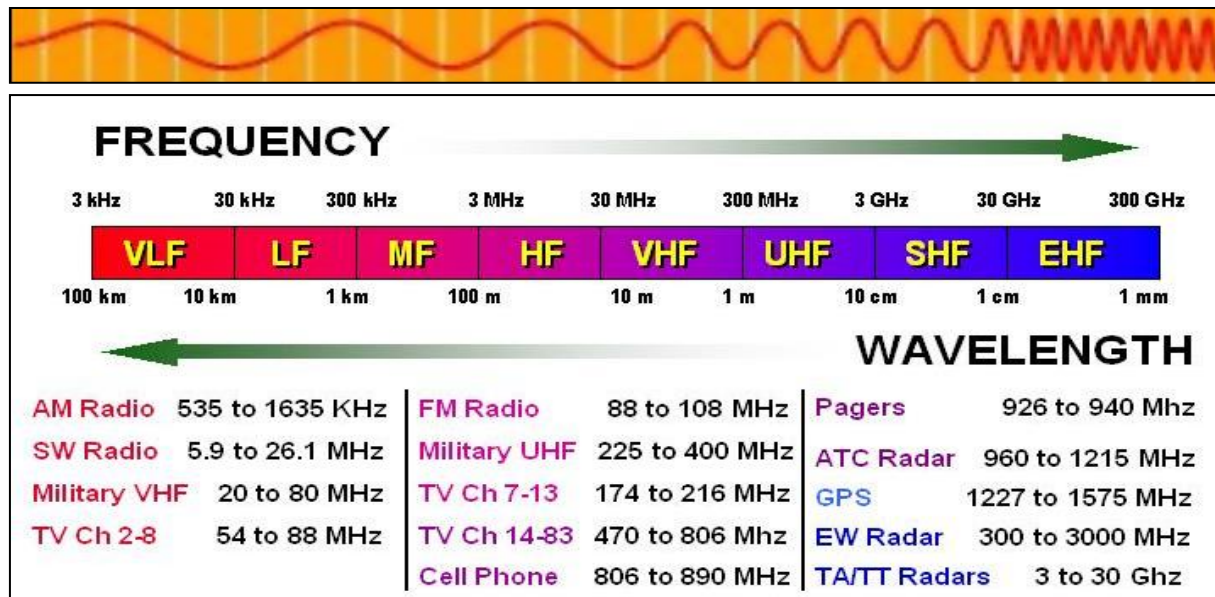
# Frequency and Wavelength



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Inversely proportional

- High Frequency = Short Wavelength
- Low Frequency = Long Wavelength



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# Electromagnetic Spectrum Bands



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A **band** within the EMS refers to grouping radiation with common properties or functions i.e.(visible light and Radio waves)

**Gamma rays:** >10EHz (Greater than 10 Exahertz)

**X-rays:** 30EHz - 30PHz (30 Exahertz - 30 Petahertz)

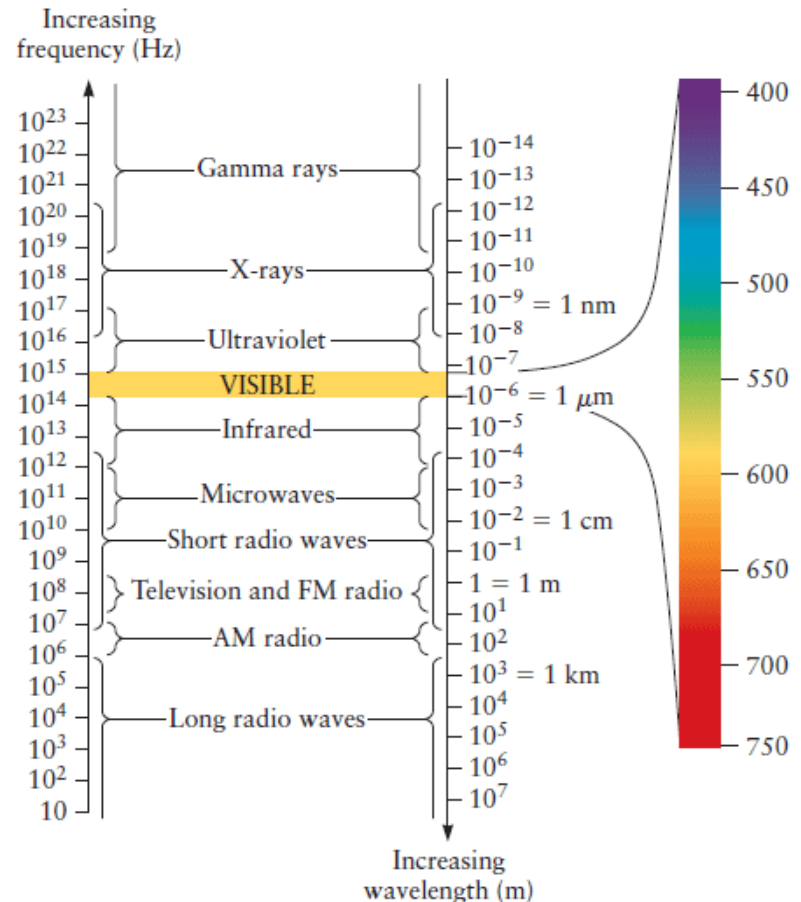
**Ultraviolet:** 30PHz - 790THz (30 Petahertz - 790 Terahertz)

**Visible Light:** 790THz - 405THz (790 Terahertz - 405 Terahertz)

**Infrared:** 405THz - 300GHz (405 Terahertz - 300 Gigahertz)

**Microwaves:** 300GHz - 300MHz (300 Gigahertz - 300 Megahertz)

**Radio Waves:** RF Band 300GHz - 3Hz (300 Gigahertz - 3 Hertz)



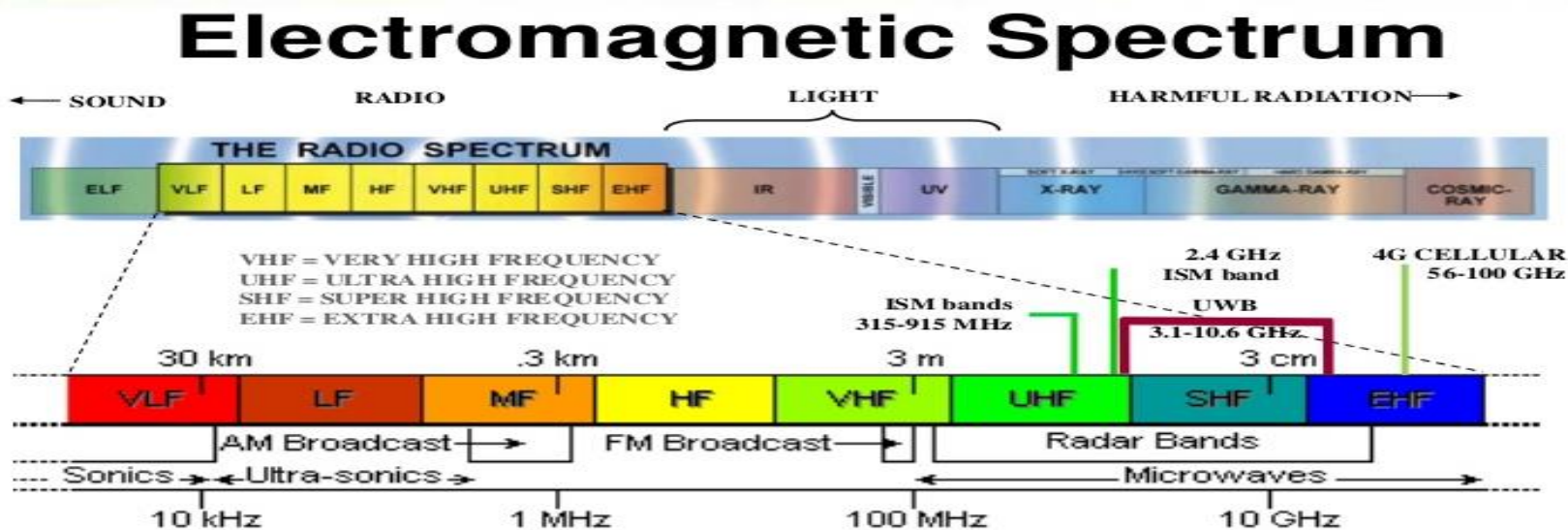
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# Radio Waves and Microwaves



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- The Radio wave band and the Microwave band encompass most types wireless communications.
- There is an **overlap** of these two bands with **no clear cutoff** between the two
- These two bands are further divided into smaller sub-bands which different types of communication systems use.



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# Questions?

# Communications Sub Bands



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| Radio Waves             |        |        |        |         |         | Microwaves |         |         |        |         |
|-------------------------|--------|--------|--------|---------|---------|------------|---------|---------|--------|---------|
| Communication Sub Bands |        |        |        |         |         |            |         |         |        |         |
| ELF                     | SLF    | ULF    | VLF    | LF      | MF      | HF         | VHF     | UHF     | SHF    | EHF     |
| 3 Hz                    | 30 Hz  | 300 Hz | 3 kHz  | 30 kHz  | 300 kHz | 3 MHz      | 30 MHz  | 300 MHz | 3 GHz  | 30 GHz  |
| 30 Hz                   | 300 Hz | 3 kHz  | 30 kHz | 300 kHz | 3 MHz   | 30 MHz     | 300 MHz | 3 GHz   | 30 GHz | 300 GHz |

Where We Operate →

**EHF** = Extremely high frequency (Microwaves)

**SHF** = Super high frequency (Microwaves)

**UHF** = Ultra high frequency (Microwaves)

**VHF** = Very high frequency

**HF** = High frequency

**MF** = Medium frequency

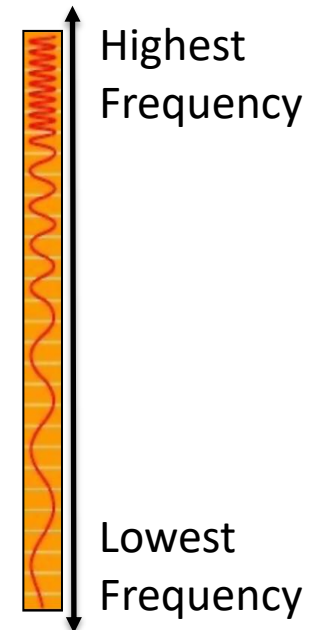
**LF** = Low frequency

**VLF** = Very low frequency

**ULF** = Ultra Low frequency

**SLF** = Super Low frequency

**ELF** = Extremely low frequency



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# Radio Waves



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Radio Waves , generally fall within the 0-300MHz portion of the EMS

- Good for transmitting small to moderate amounts of information.
- Lower portions of the band are able to cover large distance and bend around the earth or bounce off the atmosphere.
- Requires much larger antennas than microwave receivers.
- This band has been in use for communication since the late 1800s.



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# Extremely Low Frequency (ELF)

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Extremely Low Frequency (ELF) is the band of radio frequencies from 3 to 30 Hz.

- Wavelength ranges from 10,000 to 100,000 km
- Used by Russian/US Navy to communicate with submerged submarines
- Due to long wavelength, only one-way communications is possible



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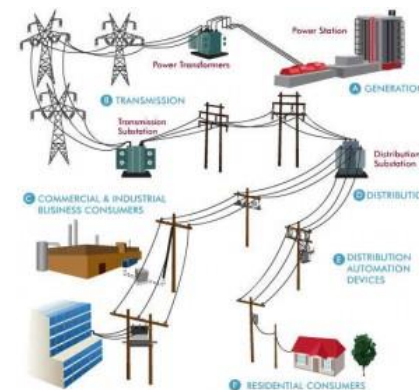
# Super Low Frequency (SLF)



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Super Low Frequency is the frequency range between 30 Hz and 300 Hz.

- Wavelength ranges from 1,000 to 10,000 km
- Range includes frequencies of AC power grids (50 Hz and 60 Hz)
- Used for submarine communications



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# Ultra-Low Frequency (ULF)



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Ultra-Low Frequency (ULF) is the frequency range between 300 Hz and 3 kHz.

- Wavelength ranges from 100 to 1,000 km
- Used for communications in mines
- Ability to penetrate the Earth
- Scientists have observed spikes in ULF waves just before large earthquakes



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# Very Low Frequency (VLF)



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Very Low Frequency (VLF) is the frequency range between 3 kHz to 30 kHz.

- Wavelength ranges from 10 to 100 km
- Can be used to communicate with submarines
- Ability to penetrate water to a depth of 10 to 40 meters
- Used for radio navigation beacons (alpha) and time signals (beta)
- Used for EM geophysical surveys



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# Low Frequency (LF)



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Low Frequency (LF) is the frequency range between 30 kHz to 300 kHz.

- Wavelength ranges from 1 to 10 km
- Used for AM broadcast service in Europe, North Africa, and Asia
- Used in the western hemisphere for:
  - Aircraft beacon
  - Navigation
  - Information
  - Weather systems



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# Medium Frequency (MF)



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Medium Frequency (MF) is the frequency range between 300 kHz to 3 MHz.

- Wavelength ranges from 100 to 1000 m
- Used for maritime distress and emergency frequency (500 kHz)
- Older cordless phones (1980's models) were within MF band



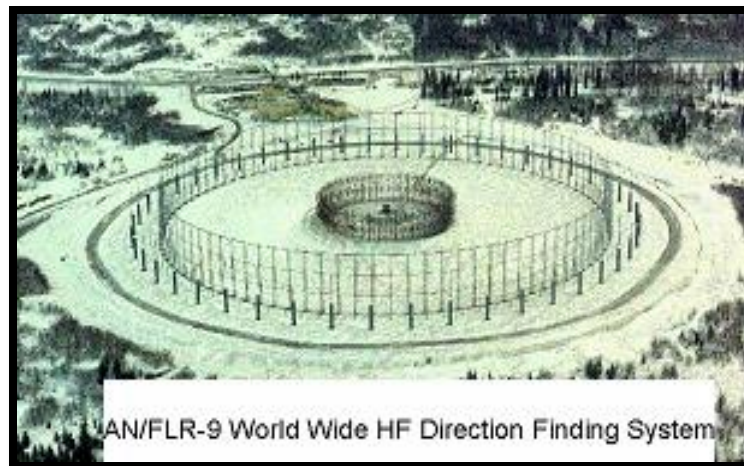
# High Frequency (HF)



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High Frequency (HF) is the frequency range between 3 MHz to 30 MHz.

- Wavelength ranges from 10 to 100 m
- Used for medium and long-distance communications
- Susceptible to interference



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# Very High Frequency (VHF)



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Very High Frequency (VHF) is the frequency range between 30 MHz to 300 MHz.

- Wavelength ranges from 1 to 10 m
- Commonly used for:
  - VHF FM radio and television broadcast stations
  - Terrestrial navigation systems
  - Maritime communications
  - Aircraft communications
- Ideal for short distance Line-of-Sight (LOS)
- Highly susceptible to tropospheric ducting



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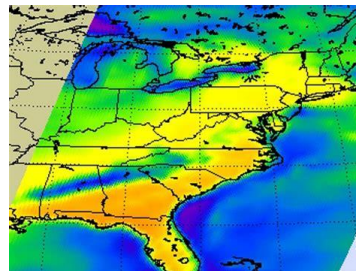
# Microwaves



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Microwaves cover roughly the 300MHz-300GHz portion of the EMS

- Good for transmitting large amounts of information.
- More affected by terrain and weather than radio waves
- Used for radar applications
- Not absorbed by most plastics, glass, or ceramics
- Widely used in modern technology



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# Ultra-High Frequency (UHF)



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Ultra-High Frequency (UHF) is the frequency range between 300 MHz to 3 GHz.

- Wavelength ranges from 1 dm to 1 m
- Commonly used for:
  - Television broadcast
  - Mobile phones
  - Two-way radios
  - Global Positioning System (GPS)
  - Wireless Local Area Network (WLAN)
  - WIFI



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# Super High Frequency (SHF)



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Super High Frequency (SHF) is the frequency range between 3 GHz and 30 GHz.

- Wavelength ranges from 1 cm to 1 dm
- Microwaves are part of this frequency
- Commonly used in:
  - Television broadcast
  - Modern radar
  - Wireless USB (Universal Serial Bus)
  - Mobile phones (W-CDMA, Wideband Code Division Multiple Access)
  - Satellite uplink/downlinks
  - Terrestrial high-speed data links (ground fixed wireless, microwave)



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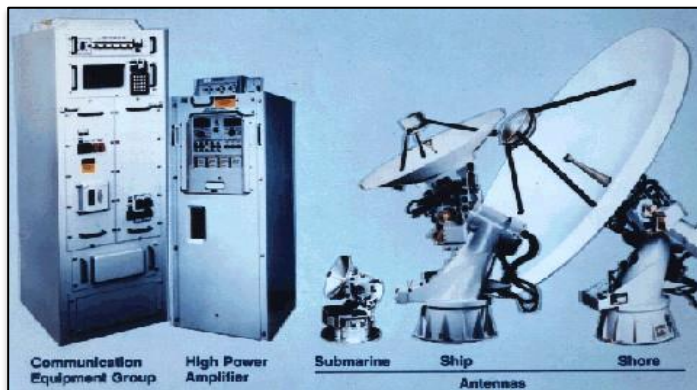
# Extremely High Frequency (EHF)



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Extremely High Frequency (EHF) is the highest radio frequency band between the frequencies of 30 GHz to 300 GHz.

- Wavelength ranges from 1 cm to 1 mm
- Extremely prone to atmospheric attenuation
- Jam resistant due to beam forming and wide band width (adaptive nulling)



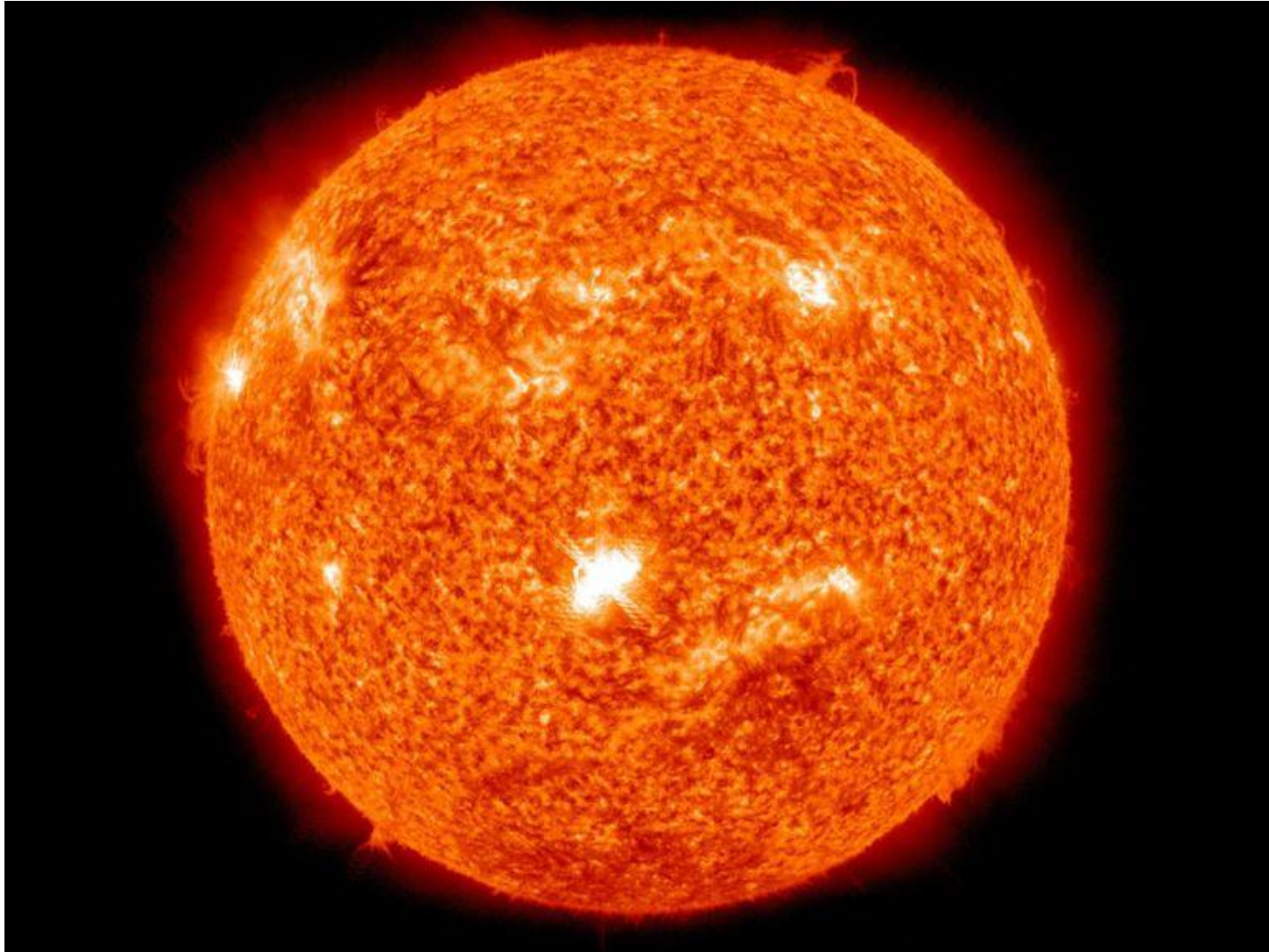
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# Questions?

# Bands above Radio/Micro Waves

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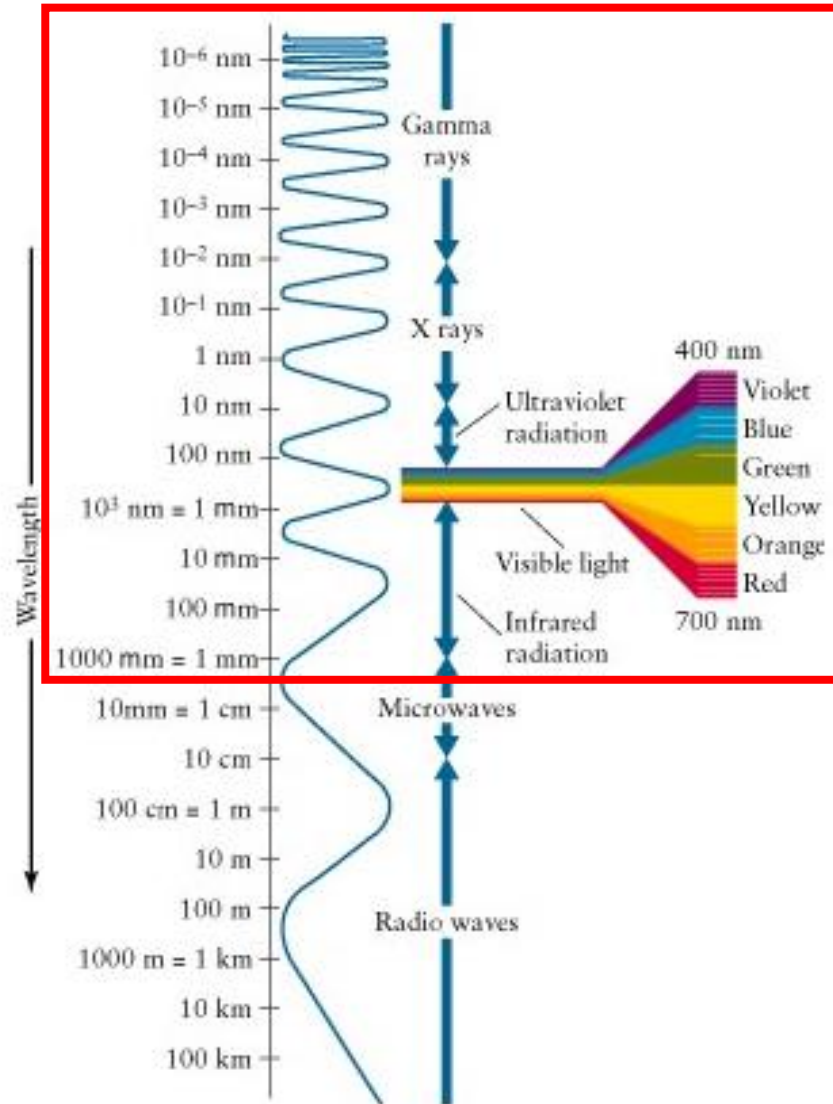


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# Placement on the Spectrum



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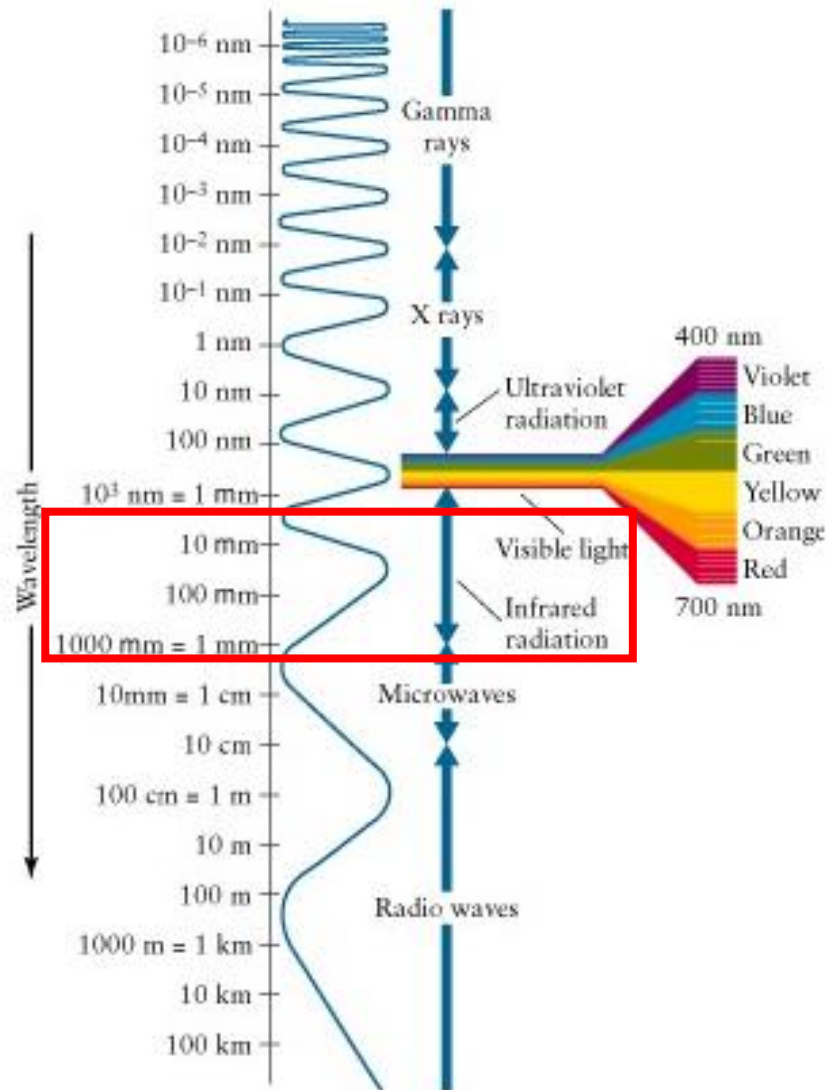


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# Infrared (1 of 3)



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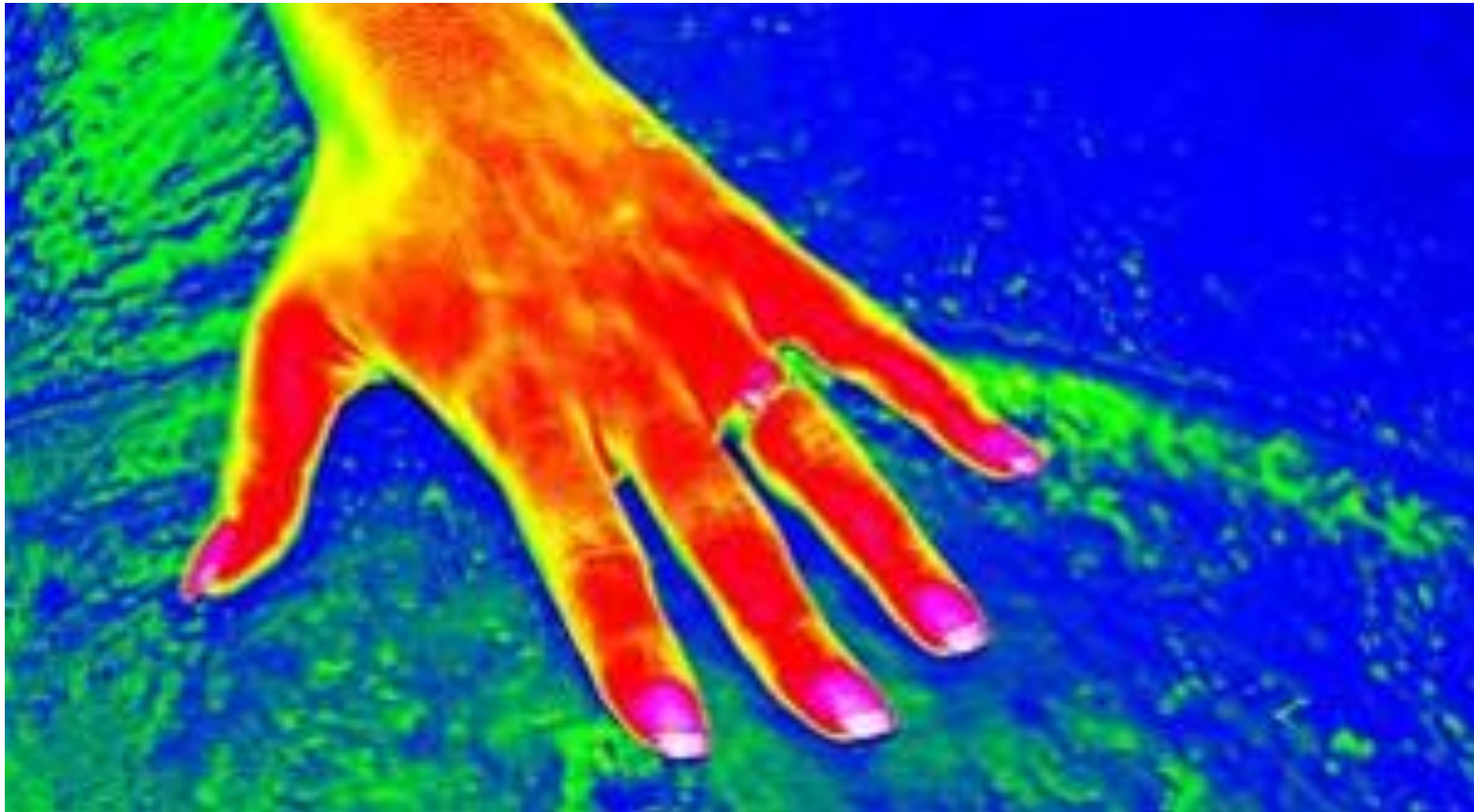
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# Infrared (2 of 3)



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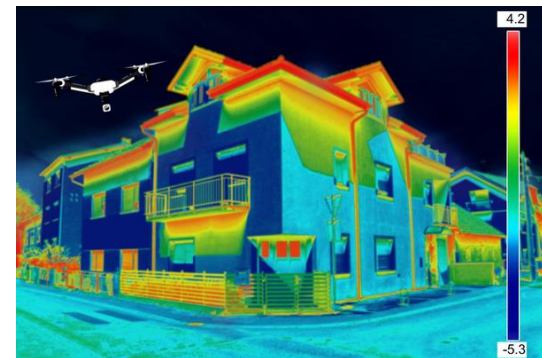
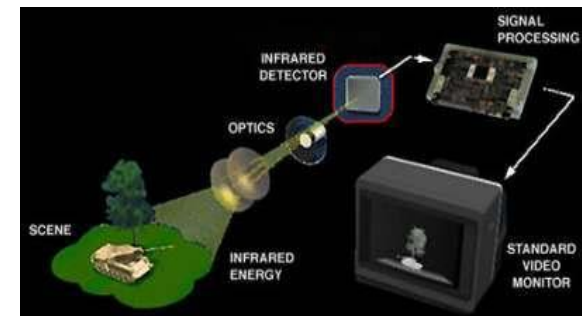
# Infrared (3 of 3)



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Infrared (IR) falls between visible light and microwave portions of the EMS with frequencies from 300 GHz to 405 THz

- Wavelengths 750 nm to 1 mm
- Near IR:
  - Close to visible light, not hot
  - Used in remote controls
- Far IR:
  - Wavelength is the size of a pin head
  - Close to microwaves
  - Thermal (Radiant heat)

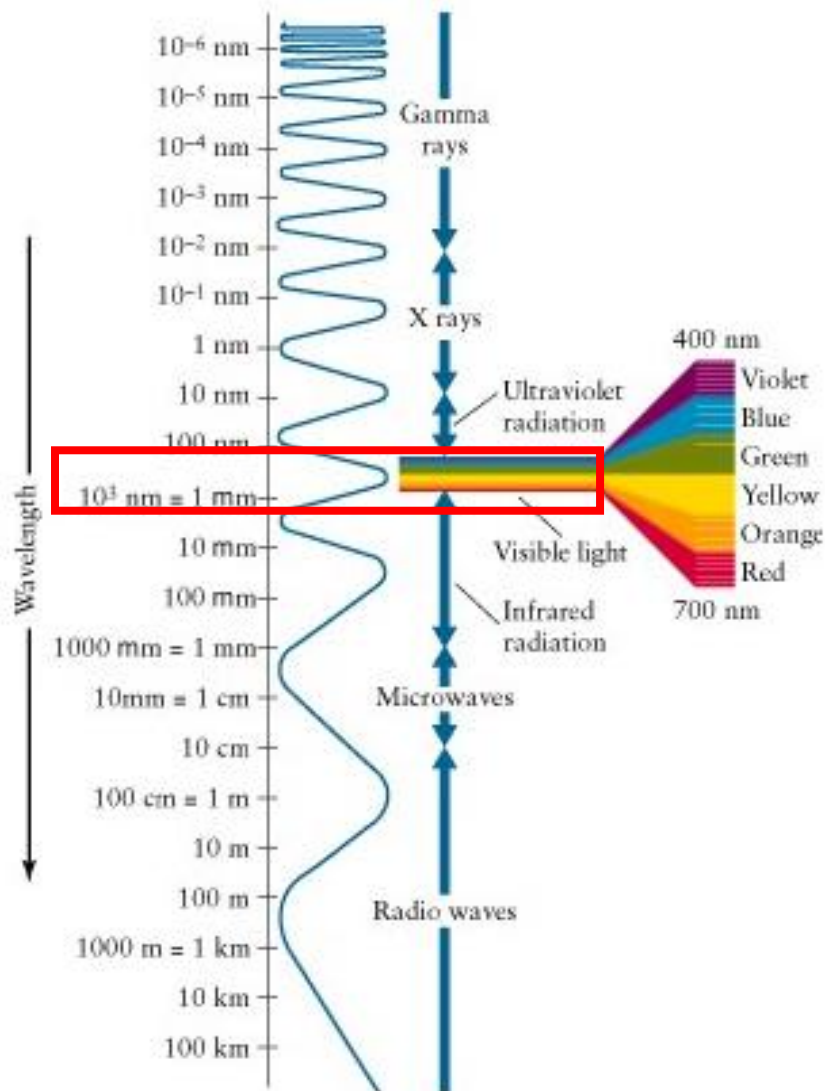


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# Visible Light (1 of 3)

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## Visible Light (2 of 3)



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# Visible Light (3 of 3)

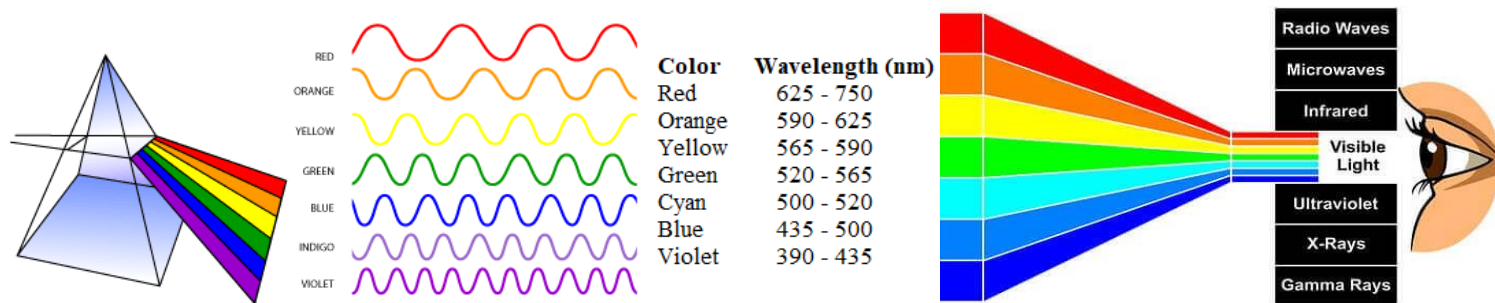


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Visible Light portion of the EMS has wavelengths capable of stimulating specialized sense organs and is perceptible as light.

Visible Lights ranges from 405 THz (Red) to 790 THz (Violet).

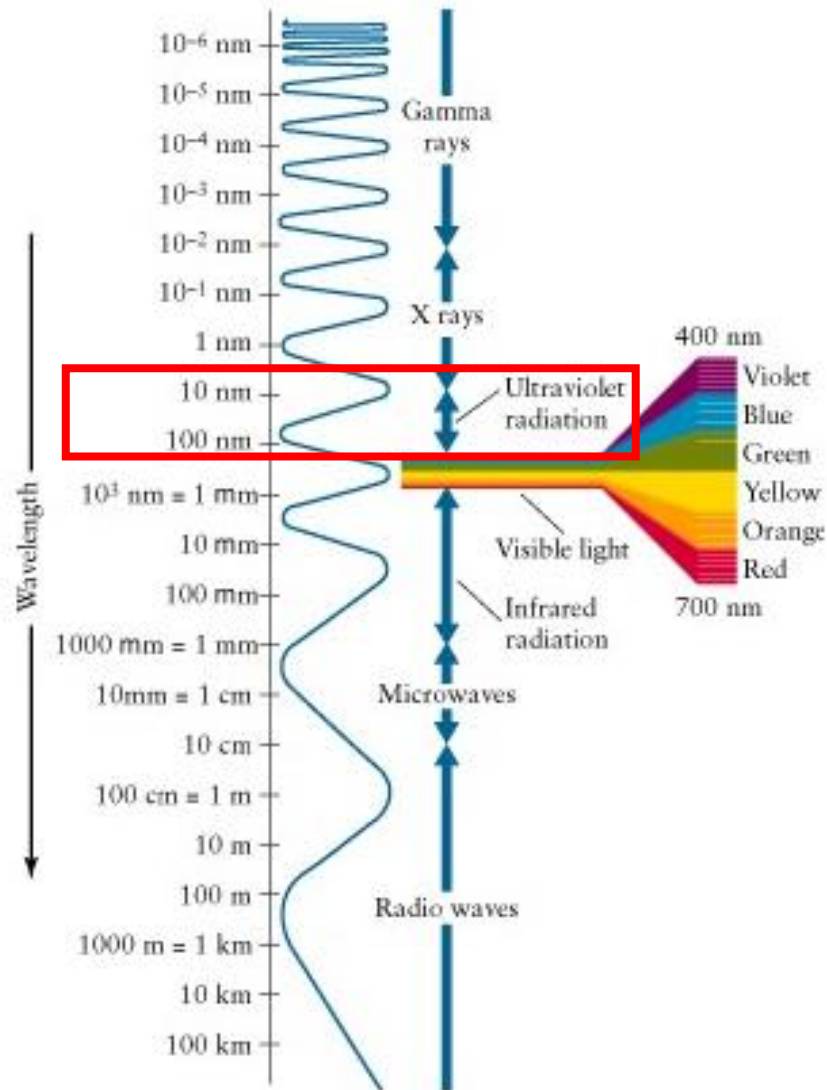
- Each color has a different wavelength



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# Ultraviolet Waves (1 of 3)

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# Ultraviolet Waves (2 of 3)



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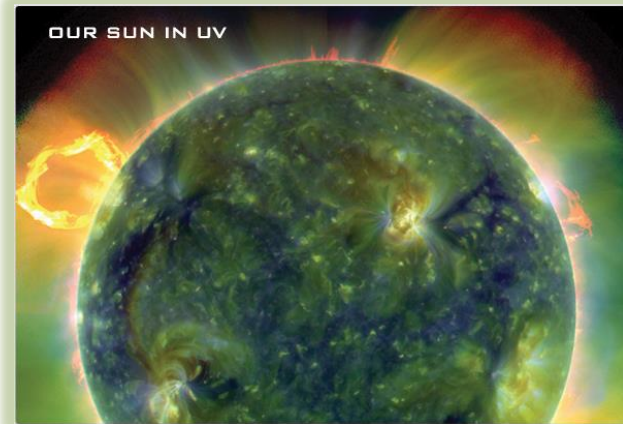
# Ultraviolet Waves (3 of 3)



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Ultraviolet (UV) light has shorter wavelengths than visible light in the frequency range between 790 TeraHertz (THz) to 30 PetaHertz (PHz).

- Wavelengths 100 nanometers (nm) to 400 nm
- Invisible to humans
- Visible to bumblebees
- Split into three regions:
  - Near Ultraviolet (NUV)
  - Far Ultraviolet (FUV)
  - Extreme Ultraviolet (EUV)
- Some UV waves are blocked by Earth's atmosphere
- Wavelengths are down to the size of cells (Dangerous)

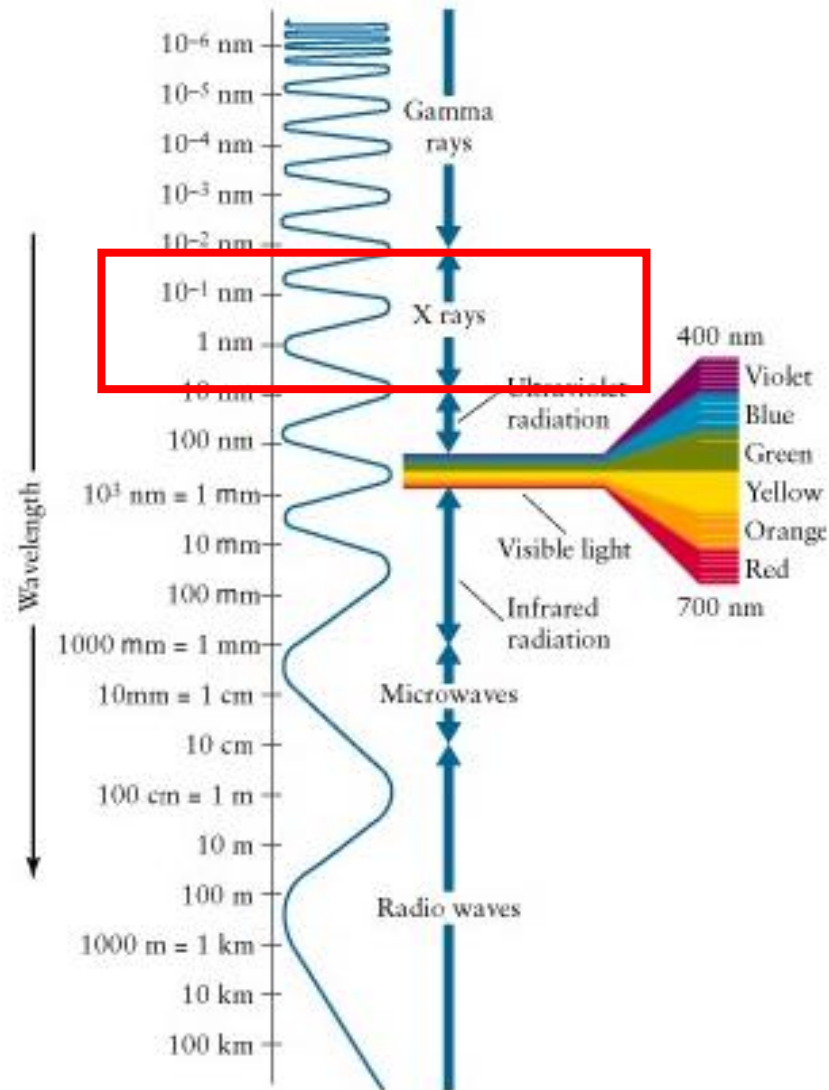


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# X-rays (1 of 3)



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## X-rays (2 of 3)



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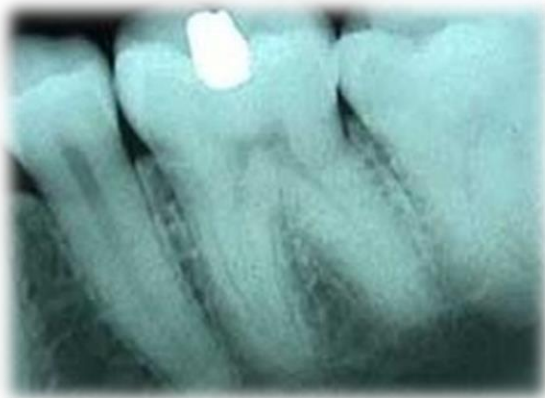
# X-rays (3 of 3)



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X-rays have frequencies in the range of 30 petahertz (PHz) to 30 exahertz (EHz).

- Wavelengths 0.01 nm to 10 nm
- Energies in the range 100 eV to 100 keV
- Able to penetrate skin and cause damage to the human body if not controlled
- X-ray detectors collect actual photons of x-ray light

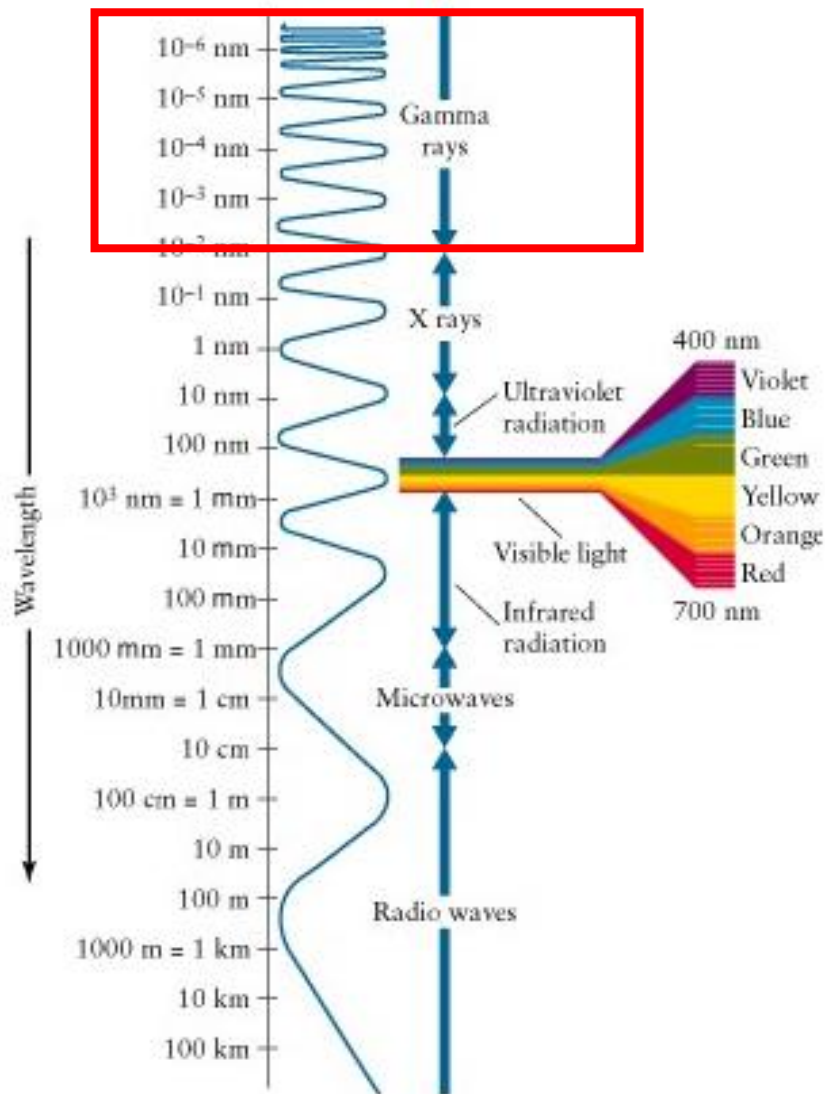


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# Gamma Rays (1 of 5)



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## Gamma Rays (2 of 5)

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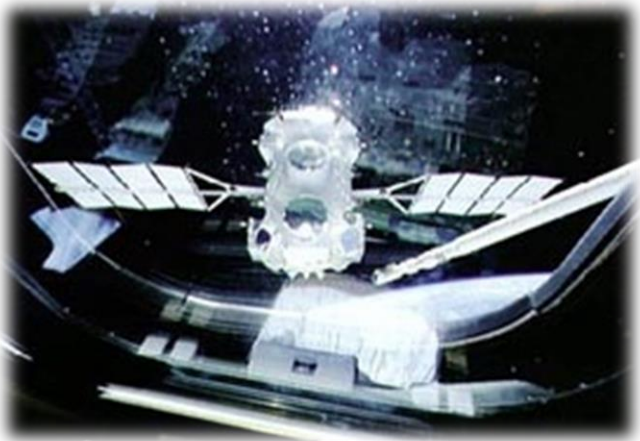
# Gamma Rays (3 of 5)



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Gamma Rays has the smallest wavelengths on the EM spectrum with frequencies of more than 10 Ehz.

- Wavelengths less than 0.02 nm
- Generated by radioactive atoms and nuclear explosions
- Ability to kill living cells (such as cancerous cells)



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# Gamma Rays (4 of 5)



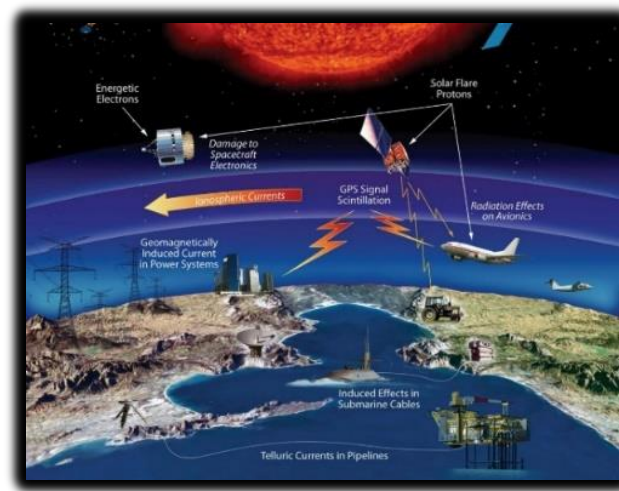
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Gamma Rays are produced by the hottest and most energetic objects in the universe, such as neutron stars and pulsars, supernova explosions, and regions around black holes.

On Earth, gamma waves are generated by nuclear explosions, lightning, and the less dramatic activity of radioactive decay.

Examples of Use :

- Medical: Treating cancer
- Industry: Testing oil pipelines
- Science: Energy and Bombs
  - Electromagnetic Pulse (EMP)



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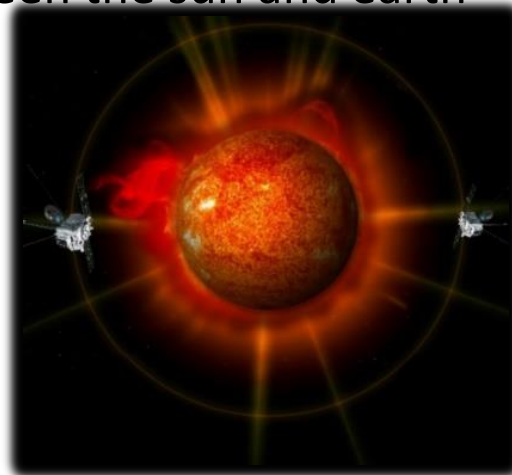
# Gamma Rays (5 of 5)



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One of the main concerns pertaining to gamma rays is Coronal Mass Ejections (CMEs) from the sun and the effect on Space Weather.

- CMEs are large expulsions of plasma and magnetic field from the Sun's corona
- Space Weather is a variation of solar phenomena in the space environment between the sun and earth



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# Questions?

# Summary



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Define the characteristics of the Electromagnetic Spectrum (EMS)

- Identify Electromagnetic Wave Characteristics
- Identify basic concepts of the Electromagnetic Spectrum (EMS)
- Identify Radio Waves Frequency Bands within the RF Spectrum
- Identify the bands above Radio Waves, and their typical uses

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# References



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- NAVEDTRA 14182. *Navy Electricity and Electronics Training Series (NEETS) Mod 10, and 17*. Sept 98.
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